

ceramic decorating

tool techniques



decorating pottery
with wax resist, slip trailers,
clay stamps, carving tools and more

Ceramic Decorating Tool Techniques

Decorating Pottery with Wax Resist, Slip Trailers, Clay Stamps, Carving Tools and More

We all love tools, especially pottery tools, and we normally think about forming when we talk about tools for ceramics, but the most useful clay tools we have, besides our hands, are tools for decorating our ceramic work. A decorative surface, of course, is the first thing noticed about a piece of pottery, and as with all things made by hand, the right tools make all the difference.

Ceramic Decorating Tool Techniques: Decorating Pottery with Wax Resist, Slip Trailers, Clay Stamps, and Carving Tools explains those tools in detail and shows you how to use them for the greatest effect for your own ceramic surfaces. Make your own ceramic colored pencils, or try using a combination of dry and wet decorating techniques to get maximum depth out of your work. For a more traditional approach that has your own personal touch, try making your own brushes!

Decorating Pottery with Clay Pencils, Wax Resist, Glaze Pens, and Slip Trailers

By Robin Hopper

There are so many ways to decorate the ceramic surface, and this overview of several decorating tools used to apply color to clay includes recipes for making your own pencils and crayons, as well as how to approach using each type of decorating tool.



Great Clay Stamps in 30 Minutes

By Virginia Cartwright

Making your own stamps is a great way to personalize and expand your pottery tool kit. Polymer clay makes it simple and fast, so you can get ideas into clay quickly.



Using Simple Tools to Decorate Ceramics with Complex Designs

By Molly Hatch

Sometimes the simplest objects are the best tools for decorating pottery. Using a pencil and laminated paper stencil, Molly Hatch walks you through how to transfer a design to a pot and then layer color onto the surface for a striking result.



Making Brushes for Pottery Decoration

By David Gamble

There is nothing more satisfying than making your own pottery decorating tools, and this step-by-step project shows you how to make personalized glaze brushes.



Bisque Texture Rollers

By Sarah Pike

Sarah Pike explains how she makes bisque-fired texture rollers and how she uses them to make her handbuilt teapots.



Decorating Pottery with Clay Pencils, Wax Resist, Glaze Pens, and Slip Trailers

By Robin Hopper

For those who are excited about the graphic possibilities of the ceramic surface and enjoy using drawing implements that have something of a sharp, scratchy or linear nature, the marks made by pencils, pens, crayons and trailers likely will make them favorite tools of expression. These tools are the foundation of written or pictographic communication in Western civilization, whereas the brush is the foundation of mark making for most Eastern civilizations. Those raised in the Western traditions usually feel more affinity with scratchy drawing tools than with the soft, calligraphic brushes. Fortunately, the range of ceramic decoration tools encompasses both soft and hard possibilities.

Ceramic Pencils

Regular pencils, with what we call “leads,” actually are made from graphite of various degrees of hardness from 6H (extremely hard) to 6B (extremely soft). Marks made with graphite pencils on ceramic surfaces will burn out in the firing, which can be very convenient, as the firing erases the guidelines or grids used for painting or draw-



Jack Sures, Canada, *Wide Bowl* (detail), ceramic ink drawing on porcelain. Private collection. Photo: Judi Dyelle.

ing on patterns and designs in ceramic pigments. Guidelines also can be painted on with vermilion watercolor paint, which also burns away.

Pencils for ceramic use (to make marks that don't burn out in firings) are made with combinations of refractory materials, clays, and colorants and are usually only commercially available in one level of hardness that would probably equate to the HB rating of a graphite pencil. HB hardness is midway between 6H and 6B. Companies that produce ceramic pencils have a habit of coming and going, but most ceramic supply houses usually will be able to find and supply them. Pencils are commercially available in a very limited variety of colors.

Ceramic pencils are normally used on bisque-fired clay that has been sufficiently hardened to withstand the pressure needed for satisfactory mark-making. Since the pencil “lead” may be quite fragile in use, the smoother the clay surface, the better the drawing.



Trailers, ceramic pens, and pencils



Verne Funk, California, USA, *Split—Portrait of the Artist*, 18 in. (46 cm) in diameter, wheel-thrown whiteware, underglaze pencil, glaze, 1996.

Bisque surfaces can be smoothed by sanding with wet and dry silicon carbide or aluminum oxide papers, or the surface of the greenware may be sprayed or brushed with a terra sigillata coating prior to the bisque firing to provide a harder working surface. Ceramic pencils may be used on the ceramic surface just like their graphite equivalent on paper. Although sharpened points tend to wear quickly on the abrasive ceramic surface, the combination of pencil tip marks, side-of-pencil marks, and the opportunity to create tones through finger-rubbing or smudging the soft image gives wide potential for drawn imagery development.

If the commercial underglaze pencils are too soft for satisfactory use, it is quite easy to make your own and harden them to a more satisfactory and less friable state. Ceramic pencil drawings can be fired onto the bisque-fired clay to harden them before glazing, or, alternatively, they

can be fired on unglazed high-fired clays, such as porcelain or stoneware, without the need for a glaze coating.

The selection of colorants or mixtures of colorants used in the coloring of the “lead” will control the effectiveness of the drawings at high temperatures, but most will tolerate cone 10.

To make ceramic pencils and pastels, use a porcelain-type slip with 50 percent white firing ball clay or plastic kaolin. For dry strength in the green state, 3 percent macaloid or 5 percent bentonite should be added.

Ceramic Pencil Slip Recipe

White-Firing Ball Clay	50 %
Potash Feldspar	25
Silica	25
	100 %
Add: Macaloid (or 5% bentonite) . . .	3 %
Colorant (maximum)	15 %

The materials, including colorants, should be dry sieved through an 80-mesh screen to ensure thorough

blending. For color, you can use mineral oxides, carbonates, and prepared stains. A variety of combinations will produce a wide range of colors, although it's important to select colorants that won't burn out at high temperatures; not many will, but cadmium/selenium and potassium dichromate are likely to do so. The amount of colorant can be up to 15 percent. More than that will cause loss of plasticity in the raw state, making it difficult to form the pencils. The more colorant used, the more intense the color.

Mix the dry materials with approximately 45 percent water, to which 1 percent of sodium silicate per 100 grams of dry material mix has been added. This will slightly deflocculate the slip, giving additional green strength while also intensifying some of the colorants.

Form the pencils by drying the colored slip to a plastic state, and then either rolling out coils or extruding lengths of the desired thickness. These then can be left as pencil lengths or cut into shorter 1–2 inch lengths. When dry, fire the pencils to between 1472°F (800°C) and 1742°F (950°C), depending on the desired hardness. A lower firing will produce softer “lead”; higher firing, harder “lead”. The short lengths can be placed in a claw grip drafting pencil (the Koh-I-Noor No. 48 drafting pencil can hold leads up to ¼ inch in diameter).

Pastels normally are used from the greenware state and are not prefired unless they prove too friable for convenient use. To make pastels, use the basic recipe above and simply form the clay into coils or extrusions to the desired size for use. If they prove too fragile, they can be fired to between 1112°F (600°C) and 1472°F (800°C) without making them excessively hard. Ceramic pastel drawings should be fired on the ceramic object to harden them before a glaze is applied; otherwise, the powdery surface likely will be spoiled in glaze application or handling. Surface powder also

might cause crawling through lack of glaze adhesion.

Crayons

To make wax crayons, mix the dry recipe above with ordinary commercial wax resist. Form the crayon, and let it dry. Since the crayon will contain some latex, it also will have a slight resist effect on the work, particularly when used on bisque-fired ware. For a crayon with greater resist qualities, stir colorants into wax, let cool, roll the wax into rods of different widths, and cut the rods in convenient lengths.

Underglaze Pens

Underglaze pens are like super-fine trailers containing an “ink” that gives good flowability for drawing. They are available commercially from a number of producers, or you can make your own with the fine trailers that are available. You can also dip any form of “nibbed” pen, from fine-pointed mapping pens, to quills or sharpened bamboo, into ceramic ink.

Black Ceramic Ink Recipe

Calcium Borate	30 %
Potash Feldspar	30
Ball Clay	25
Silica	15
	100 %

Add: Bentonite	5 %
Mason Stain 6600	
or other black stain	10 %

Thoroughly dry-mix these ingredients, then add a mixture of water and 5 percent sodium silicate (100 milliliters water to 5 grams sodium silicate). Pass it through a 100-mesh sieve twice. Thin the ink as appropriate for your use. This ink should work at all temperatures up to cone 12. It can be thinned to produce pen and wash-like drawings or used with a ceramic watercolor or glazes. Other colorants also can be used with this base.

Ceramic Watercolor Recipe

White-Firing Ball Clay	50 %
Potash Feldspar	25
Silica	25
	100 %

Add: Macaloid (or 5% bentonite) . . .	3 %
Colorant (maximum)	15 %

For watercolors, the materials are mixed together, then enough water is added to make a slip, which is passed through an 80-mesh sieve and poured onto a plaster surface. When dry to the touch, watercolor cakes can be made by forming rounds or squares of the colored slip and letting them dry completely. They then can be used like ordinary children’s watercolors by wetting the surface with water and applying with a brush.

Trailers

A wide range of trailers for slip, ink, glaze or overglaze uses are available from ceramic suppliers, kitchen stores, and drugstores. They usually consist of a rubber or neoprene bulb or container and a nozzle with a fine-aperture tip, or sometimes multiple tips. The simplest to find is usually either a hair coloring applicator bottle or a child’s nasal bulb aspirator from a drugstore.

Ceramic suppliers often have fine-tipped trailers, sometimes with interchangeable tips of differing aperture. The aperture of the tip required depends on the thickness of the material being squeezed through. Thin inks will go through a fine tip without clogging, but a wide tip may be needed for slips or glazes to flow properly.

As with any tools, you’ll need to practice to get the correct “feel” to achieve the best results. Keep a thin needle tool nearby when working with trailers, because the fine ones tend to clog quite easily.

This article was excerpted from Robin Hopper’s Making Marks published by The American Ceramic Society.



Lynda Katz, USA, *Bayou Boogie Wookie*, 13 in. (33 cm) in height, thrown and faceted porcelain, underglaze pencil drawing with luster glazes, 1984.



Lynda Katz, USA, *Covered Jar*, 8 in. (20 cm) in height, thrown, altered, and hand-built porcelain, glaze-trailed decoration, 1997.

Great Clay Stamps in 30 Minutes

by Virginia Cartwright



Polymer clay can be used to create a variety of clay stamps in a short period of time.

Stamps enhance your work by adding interesting textures to your pieces and depth to your glazes. For years, I carved stamps from small plaster blocks or from leather-hard clay that was then bisque fired before use. I wanted a way to make stamps more quickly so I could share them with my students and workshop participants and found that polymer clay (available at most craft stores) makes a clean, crisp impression that can be cured and ready for use in almost no time at all.

Supplies

There are several brands of polymer clay available (Sculpey, Fimo or Premo), all of which are good. Sculpey has a product called "Super Sculpey" which I use because it is a strong, shatter-resistant material. Super Sculpey is only available in a tan color, so I combine it with about 25% of another color of polymer clay if I want to change the color.

You'll need a clean, non-porous surface to work on, and a Plexiglas roller made just for polymer. You can substitute this roller with an 8-inch piece of plastic pipe. You should also buy a 6-inch long cutting blade and an inexpensive pasta machine (both available at craft stores or garage sales). The pasta machine is used to soften the clay and to blend the colored clays together.

If you're just making a few stamps, you can use your hands and roll the clay into coils until it softens. I find it helpful to attach handles to the stamps. Buy a bottle of liquid polymer clay and use this as glue to attach the polymer handle to the top of the stamp.

When making a stoneware stamp using regular clay, you need to let it dry then bisque fire it before you can put it to use, which could take several days to a week. Polymer stamps, by comparison, are ready for use in about 30 minutes. You can then take those stamps, press them into another piece of polymer clay, and quickly get a negative version of your designs.

Polymer clay does not crack or crumble as easily as stoneware or earthenware clays, and scraps can be easily recycled. Since it's not water based, the clay doesn't dry out. This property also makes polymer clay an ideal material for making impressions from a variety of objects, including antique furniture, kitchen tools, buttons, Indian wood blocks and tombstones.

As I began to explore the possibilities of this material further, I discovered that I could use my inlaid colored clay techniques to make the stamps beautiful as well as functional. By layering and blending colors, I can create an endless variety of intricate patterns.

The Process

Condition the polymer clay by rolling it ten times through a pasta maker on the thickest setting (*figure 1*). If you do not have a pasta machine, manually roll the clay into coils in your hands. Your body heat will soften the clay. If you want to mix your own colors, make coils of two or three colors, roll them into one coil and twist



Polymer clay stamps are best when used on slightly firm slabs of ceramic clay. If the stamp sticks to the clay during use, baby powder or cornstarch can be used as a release agent. These embellished slabs can be used to make handbuilt vessels and sculptures, or left flat for tile work.

the coil like a candy cane. Cut the twisted coil in half, join the two pieces and twist them again, repeating this process until the colors are blended.

You can make stamps using several layers of polymer clay by first rolling it out into a slab that is about ¼ inch thick. Thin polymer clay slabs take a deeper, clearer impression than a thick one. Put a pinch of baby powder or cornstarch on one side of the clay. Next, press the clay (powder side down) on top of a textured object. Place the polymer clay over the texture and press it with your thumbs, rather than pressing the texture down onto the polymer clay (*figure 2*).

Leave the polymer clay on the textured surface and add a second layer of Super Sculpey clay (*figure 3*). Press the layers again with your thumbs. Remove the clay and trim the edges.

Bake the polymer in a small toaster oven, following the directions given on the package. Typically,

it will bake at 275°F for fifteen minutes, though you may increase the time by five minutes if the stamps are very thick. Be careful not to overheat the polymer clay. You can cover your pieces with aluminum foil to prevent scorching and blistering (*figure 4*). If you smell a strong odor while cooking the stamps, it means that they are getting too hot. Turn off the oven, and ventilate the room.

Note

These stamps are not as strong as ceramic or wooden stamps. If you have trouble with your stamp breaking, try making the layers of the stamp thicker.

Finally, add a top decorative layer and a handle. I join everything together with a thin coat of liquid polymer clay and cure the piece again. The heat fuses the layers and handle together.

Virginia Cartwright has taught ceramics for 22 years, and has shown her work in numerous exhibitions. She currently lives in Fallbrook, California.

Tip

If you plan to make a lot of pieces, buy a toaster oven dedicated just to curing polymer pieces.



Using Simple Tools to Decorate Ceramics with Complex Designs

By Molly Hatch

I have always been interested in drawing. As an undergraduate, I focused on drawing for the majority of my time in school. It wasn't until my final year that I was shown surface decoration techniques for clay that are similar to printmaking and drawing processes. It was this marriage of drawing and clay that has driven the development of my current work.

There's something magical in the ability to interpret what I see through my hands. I think of drawing as a visual language similar to writing; both can be communication tools. I am often surprised by the small narratives that appear in the patterns I draw on the surfaces of my pots. Each bird has its own distinct personality and expression . . . A moth will buzz around a peony. The patterns I draw are always my interpretation and representation of an already existing pattern. I sometimes combine elements of different patterns, in turn creating new patterns. I play with the scale of the pattern on the pot. How the pot frames the image often dictates the pattern itself.

I spend a large amount of time looking at historic fabrics as source material and I'm always collecting new patterns to add to my repertoire. I pull out new patterns when I need a challenge and I draw the pattern on paper a few times to familiarize myself with it before experimenting on my pots. I use porcelain for my work for its durability and translucence. I love the similarity of pure white porcelain to a blank piece of paper. My forms are inspired by contemporary product design, 18th century European factory ceramics as well as the English ceramics of the Leach/Cardew studio tradition.



Finch Cup with Baroque Frame,
13 inches (33 cm) in width.

Throwing a Blank Canvas

When I'm throwing, I think of the pots and their forms as that blank piece of paper. I strive to keep my forms simple, quiet, and uncomplicated. This simplicity allows the drawing to become the major focus of each pot, rather than a competition between form and surface.

The simplest and most popular pot I make is a tumbler form that I refer to as a beaker. I use about a pound of clay to make the beakers pictured in this article. When throwing, I use very few tools other than



1 When throwing, use the crook of your finger to shape the lip of a tumbler.



2 Laminated paper template of drawing can help maintain consistency in a design when transferring images to a set.



3 Gently wrap the laminated pattern around the cup and use a quill or pencil to trace the image.



4 Remove the template to reveal the transferred tracing image now impressed into the clay.



5 Use the transferred image as a guide for drawing deeper lines into the surface.



6 Finish off the rest of the drawing freehand, using the template as a visual reference.

my hands. I use the crook of my forefinger and middle finger to shape the lips of my pots. When I do use tools, my favorite rib is a square rib fashioned after a Michael Cardew design and made for me by my husband who is a woodworker (*figure 1*). I'm careful to use the rib as little as possible because I enjoy the pots much more when there's evidence of my hand in them. When I'm trimming, I usually use a bat dampened slightly with a sponge. I tap the pot I'm trimming on center and then, using the slightly damp surface of the bat, I apply pressure to the base of the pot, which creates a slight suction and secures the pot to wheel

for trimming. When using this trimming technique, it helps to keep one hand on the pot at all times to catch it in the event that the suction gives way.

Image Transfer

Mishima is a traditional Korean slip-inlay technique. The Korean pots you see with mishima decoration typically use several colors of slip inlaid into the same piece. I basically use the same black slip recipe for all of my mishima drawing. I always refer to a pattern when I'm drawing on my pots and sometimes use a template to transfer a detail of the pattern (*figure 2*).



7
Apply a layer of stained slip over the drawing using a wide brush.



8
Wipe the excess slip from the surface of the pot using a clean sponge.



9
Use a vitreous engobe mixed with a brushing medium to create color accents.



10
Though it fluxes a bit at higher temperatures, the engobe can be used to fill in color areas on the bottom.



Tumbler, 5³/₄ inches (15 cm) in height, porcelain, mishima slip inlay, vitreous engobe, fired to cone 6.

In this case, I am using the template to transfer the bird in the pattern onto the cup surface. I make my templates by laminating my own drawing of a found pattern. This is helpful if you are trying to make multiples, but still requires a lot of drawing and interpretation because you're drawing on a three-dimensional surface.

Inlaying the Color

All of my mishima is done when the pots are a firm leather hard. Usually they are ready to draw on just after trimming is finished. To follow this technique, start the transfer by gently wrapping the laminated

Drawing Tools

There are many tools you can use to incise the surface of the pot for mishima. I have gone through stages of preferring particular tools—pencil-style X-Acto knives, commercial stylus carving tools (sold in ceramic supply stores), African porcupine quills (available at Santa Fe Clay) among others. My current drawing tool of choice is a calligraphy pen with interchangeable metal tips. It's the same kind of pen that you dip in ink and would use to do traditional calligraphy; I just use it on clay instead.

pattern around the cup, taking care to position the image exactly where you would like it to be on the cup. Then, using an African porcupine quill (dull-tipped pencils work well too), transfer the image by tracing over the lines on the template with enough pressure to draw into the surface of the pot underneath (*figure 3*). Remove the template (*figure 4*) and use the transferred image as a guide for drawing deeper lines into the surface of the pot (*figure 5*). You don't need to draw very deeply into the surface for mishima to work. I often feel as though I am just scratching into the surface of the clay. After going over the tracing, finish off the rest of the drawing freehand, using the template pattern as a visual reference (*figure 6*).

Just before you apply the slip to the surface of the pot, use a soft-bristled brush, such as a shaving brush, to get rid of as many crumbs as possible. If you don't remove the crumbs, they can often stick to the pot and create problems when removing excess colored slip from the surface later.

Using a wide brush, apply a layer of stained slip to the drawing (*figure 7*). I tend to use a thinner slip so that it has an easier time getting into the small details of the drawing. Thicker slips tend to only partially fill in the drawn lines in the surface. Once the pot has dried back to the dry leather-hard state and any sheen on the slip has disappeared, begin to wipe away the excess slip from the surface of the pot using a clean sponge (*figure 8*). You need to clean the sponge often during this process to avoid leaving streaks of color on the surface of the pot. As you work, you'll see your design emerge. The slip remains in the recessed lines of the drawing, creating the inlaid design. Tip: If you are using a clay with a lot of grog, it's a good idea to alternate between using the sponge to remove the slip and smoothing the surface with a rubber or metal rib.

At this point let the pot dry completely and then bisque fire it.

Adding Color

On many of my pots, I add color accents to the mishima pattern through painting. I do all of my painting after the pot has been bisque fired and before I do any glazing. For the color, I use a cone 04 vitreous engobe that I mix myself, but commercial underglazes also work well. If you use an engobe, combine it in a 1:1 ratio with brushing medium using a palette knife until it is well mixed (*figure 9*). The mixing-medium helps make the engobe more brushable and thins it out so that you can build up color in layers, similar to painting on canvas. This layering makes for more solid colors with less visible brush strokes.

The engobe recipe that I use tends to flux a bit at cone six but it can still be used to fill in the line drawings on the bottoms of pots (*figure 10*). After I finish adding the color, I use a clear glaze over everything except the bottom of the pot, then fire the work in oxidation to a hot cone six.

Molly Hatch is a full-time studio potter in Northampton, Massachusetts. She studied ceramics as an undergraduate student at the School of the Museum of Fine Arts in Boston and received her MFA from the University of Colorado, Boulder. To see more of Hatch's work, visit her website at www.mollyhatch.com.



Molly Hatch working in her studio while she was a resident at the John Michael Kohler Arts Center, Sheboygan, Wisconsin.

Photo Credit: Jeff Machtig. Courtesy of the John Michael Kohler Arts Center.

Recipes

Andrew Martin's Brushing Slip

(up to Cone 10)

Ferro Frit 3110.....	30 %
Ball Clay	20
Mason Stain 6600 (black)	50
	<u>100%</u>

Add: CMC

This is the slip I use for the mishima inlay areas on my work. Be sure to mix it thin enough that it fills in all of the fine lines. If using a coloring oxide (like iron oxide, manganese dioxide, chrome oxide, cobalt oxide, cobalt carbonate or copper carbonate, for example) instead of a commercial stain, the amount of colorant needed may be less than in the above recipe.

Vitreous Engobe

(Cone 04–6)

Talc.....	15.3%
Ferro Frit 3110.....	18.4
Kentucky OM4 Ball Clay.....	15.3
EPK Kaolin.....	5.1
Glomax (Calcined Kaolin).....	25.5
Silica.....	20.4
	<u>100.0%</u>

Add: CMC.....

Macaloid

Add stains to the above base at a ratio of 1:1. I use this on bisque ware.

Brushing Medium

To make a brushing medium for use with the Vitreous Engobe, slake a 50/50 mix of Macaloid and CMC in hot water and blend together until smooth. To combine the engobe with the brushing medium, start with a small amount of each and use a palette knife to mix them together. Add more medium or engobe until you get the right consistency for brushing.

Val Cushing Clay Body

(Cone 6)

Nepheline Syenite.....	23 %
EPK Kaolin	35
Tile 6 Kaolin	15
XX Sagger Ball Clay.....	5
Silica.....	22
	<u>100 %</u>

Add: Bentonite

I use a commercial cone 6 porcelain from Sheffield Ceramic Supply, however, this Val Cushing cone 6 body is great.

Making Brushes

by David L. Gamble

Using the right tool for the job is always a good idea, but in ceramics, using the right brush is critical for creating certain marks on your pots. You'd have a difficult time trying to use a short flat brush to create a long thin line, yet when using the right brush, you can create that line without any effort at all.

Brushes are made from a wide variety of either animal hair or synthetics. The material affects how the brush loads and disperses a medium. For example, red sable hair is the best choice for watercolors and washes even though less expensive brushes can be made from camel, squirrel or horse hair. Stiff boar or hog bristles are good for oil paint, while in ceramics we tend to use hairs that load a lot of glaze, so we choose goat and China bristles (the industry term for hog hair). China bristles are normally longer and stiffer than goat hair.

By making your own brushes, you can control the qualities that are important and unique to you. Though manufactured brushes are made to perform well with certain media, as artists we tend to use whatever will work to make the marks we want.

I've been constructing brushes on and off for many decades, and have presented workshops over the years, first along with potter Steve Howell, and most recently with my wife Tracy who is also a ceramic artist. We have participants make brushes on the first day of our two-day presentations, and use them on the second day. During one demonstration at Atlantic Pottery in Florida last year, Bob Kirk, a high school art teacher, was demonstrating making brushes and told me he has made brushes in classes with his students. I think that everyone from students (at all levels) to seasoned artists can appreciate the quality of a good brush after spending the time making their own.



Your surface decoration is one of a kind. So why not make brushes customized to exactly fit your techniques and style?



From left to right: Gray squirrel, deer (natural), deer (dyed red), and red squirrel.

MATERIALS LIST

- Bamboo: Select pieces $\frac{1}{4}$ – $\frac{1}{2}$ inches in diameter. There are more than 1450 species with 450 species sold in the US, so there's a wide variety available! If you cut fresh bamboo, allow it to dry for a month before using for handles.
- Hair: I've found that deer and squirrel tail hair work well to hold and disperse the glaze. Squirrel and deer tails can be purchased at any fishing store that sells fly fishing materials to tie flies. Deer tail is packaged natural and in a variety of bright colors to imitate the color of insects. Also try dog, skunk, elk, fox, goat, etc. Note: Human hair is too limp to use.
- Dental floss: Use strong dental floss to tie the hairs together. Waxed dental floss is a bit more sticky and easier to tie. Non-waxed dental floss is a bit harder to tie but accepts the glue better. I prefer the waxed dental floss.
- Masking tape
- Sandpaper
- Hacksaw with fine blade
- White glue
- Needle tool
- Small drill bit ($\frac{1}{8}$ – $\frac{5}{32}$ inch in diameter)
- Cotton thread/string



1 Tie the base of the hair with floss.



2 Trim the hairs at the base.



3 Dip the cut end in glue.



4 Wrap and tightly tie the rest of the floss.

Process

To begin making your brush, select the hair you're going to use and get an extra long piece of dental floss to wrap the bottom of the hairs. Four hands work best for this so get someone to hold the hairs as you tie them (*figure 1*). Do not trim the floss yet. Next, cut the hairs at the base. Start with at least one inch of hair. If you want a longer brush, deer tail hair can be 3–4 inches in length or longer if you use most of the tail. Remember that you'll need to leave about ¼- to ½-inch of hair to glue and wrap together (*figure 2*), so if you want a 1-inch-long brush tip,

cut the hair to 1½ inch. Dip the cut end of the brush hair into a puddle of white glue (*figure 3*). Allow the hair to soak up the glue. Wrap and tightly tie the rest of the floss, forcing the base of the hairs together into a tightly bound tube-like shape (*figure 4*). Allow to dry before gluing into the handle.

Choose a piece of bamboo that your brush hair will fit into tightly. Hold the bamboo behind a node like a brush to make sure it's comfortable. Bamboo is hollow between the nodes so decide how long you want the handle, and leave about ¾ of an inch before you reach the node to create a natural ferrule you can fill with glue.



5 Cut bamboo with a fine tooth saw.



6 Sand both handle ends smooth.



7 Test fit the brush then add glue.



8 Insert brush then use a needle tool to shove it in tight.



9 Drill a hole in the top of the handle for hanging.



10 Split the bamboo end with a knife.

Wrap masking tape around the end of the bamboo where you plan to cut and use a fine-toothed hacksaw or jeweler's saw to cut the bamboo (*figure 5*). The tape keeps the bamboo from splintering. Sand both ends smooth (*figure 6*).

Test fit the brush hair in the handle then fill the ferrule space with glue (*figure 7*) and force the brush hair into it. Use a needle tool to shove it in tight so that the dental floss wrapping cannot be seen (*figure 8*). Let it dry.

Drill a hole through the bamboo at the end of the handle using a small drill bit so you can add a thread to hang the brush up (*figure 9*). You can also sand flat a section at the top of the handle so you can write your name on the brush with a permanent marker. Tung oil can be used on the handle for a finishing touch. No other finishing or waterproofing is needed. Over time, the oils from your hand will give the brush handle a nice patina.

As an alternative, you can also split the bamboo ferrule with an X-Acto knife deep enough for the brush

hair to fit in (*figure 10*). This works well if the opening in the bamboo is a bit too small for the brush hair to fit in. Place glue inside and insert the brush hair. Wrap the end tightly (*figure 11*) then place the end of the thread through the loop you have left at the top and pull the bottom thread until the end is pulled into the wrapping (*figure 12*). Trim off excess thread and add a thin amount of white glue over the wrap. I've also heard of thin copper wire being used.

With either method, once you've secured the brush hairs into the ferrule and the glue has dried, thread a piece of ribbon, twine, string or a leather strap through the hole at the end to create a hanging loop (*figure 13*). This finishing touch is just one more way to create your own special brushes.

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Wrap the end with thread or wire, creating a loop with the beginning of the thread and wrapping around it to secure.



Place the end of the thread through the loop and pull the bottom end of the thread down to pull the end through.



Thread a cord for hanging the brushes through the hole drilled at the end of the handle.

Bisque Texture Rollers

by Sarah Pike



Twiggy Cream and Sugar, mid-range red stoneware, gold glaze.

Teapots are one of my favorite things to make. They always have been. These teapots are slab built with eighteen separate pieces. That's a lot of pieces to play around with, and playing around with them is what gives the teapot its personality and makes me a happy potter. I like all those parts to be evident too—either sharpened with a rib or defined with a wooden tool. The seams become the lines that define a change in plane and become the drawing that defines the shape of the pot. I also like what a teapot implies: A social event where you get to share tea with others. A mug is more intimate and I love mugs too, but a teapot symbolizes getting together, getting warm and comfortable, and having a good chat.

With complicated forms like teapots, I find that three is the magic number to make at the same time. It allows one part to set up while I work on another, but it also allows for play because, with more parts, I can interchange them between forms to get the look I want. If I try to make more than three, I lose engagement with each individual pot and that time and attention to each pot is important to me.

Soft Slab Building

I use soft slabs because the finished pot ends up with a soft quirkiness that appeals to me. Slabs should be soft enough to wedge back up again, but stiff enough that they won't collapse. If they're too stiff, I get cracking or rigidity in my work. I use a commercial, finely sieved, mid-range red stoneware. The result is that the unglazed areas are rich in color due to the high red-iron content, but satiny soft to the touch because the clay is so fine. At the beginning of my day I roll out a stack of ¼-inch-thick slabs and smooth out unwanted textures with a soft rib. It's always more fun to make slab work if you have a pile ready for your next step.

Texture

I love texture, the way glaze breaks over it and the feel of it when I am holding a textured pot. It's one of the reasons I transitioned from throwing to hand-building. I make my own stamps and texture rollers (figure 1), which lends an idiosyncratic, handmade, quirky appeal and imbues personality into my work.

My texture rollers are simply ceramic cylinders with various lines, dots, textures, or stamps pushed into them, then bisque fired. They are super fun, simple, and addictive to make. My stamps are more time-consuming as they are meticulously carved out of leather-hard clay with tiny trimming tools. The tricky part is deciding which areas will be positive and negative because the end imprint will be reversed. Which parts do I want raised for the glaze to break over or imprinted for the glaze to pool into?

The Body: Texture and Shape

Transfer a large slab onto cement board (such as HardieBacker® Cement Board) (figure A), and roll or stamp in texture (figure 2).

Measure and cut two rectangles for each teapot. The dimensions of the rectangle really affect the personality of the pot, so I like to play around with these dimensions. These slabs are around 4½ inches (12 cm) tall by 10 inches (26 cm) wide. Always over bend a slab to give it that bendy memory and avoid tension at the seams because slabs like to open up as they dry. For this I use a large-diameter pipe and simply drape the slab over the curve (figure 3). (Note: I use the slabs immediately, while they're still soft. I just use the pipe to create a bend. If I were to make the curve without the pipe, the slab would collapse.) Stand the two curved slabs upright and score and slip the end seams together (figure 4). Puff out the walls by gently sliding your fingers up the inside for extra volume.

Cut two leaf shapes to form the top. Score, slip, and compress the joins with a rib or brayer being careful not to damage the texture on the sides (figure 5).

The Collar and Lid

I make the collar out of a short and wide cylinder, squeezed into an oval, sometimes with more texture or another texture altogether (figure 6). Before joining the seam, I soften the cut edge by gently rolling a brayer at a 45° angle along the cylinder's length. This gives the rim a finished look. Cut the base of the collar to fit the undulating top form of the teapot. Add a lid seat with another narrow slab then attach. So many bits!

Okay, now for the lid. For this you'll need leaf shapes again, but this time, little round leaves. Make another small cylinder with the top edge cut at 45° and bottom edge softened for that finished look. Join this cylinder to

Cement Board

Switching from drywall to HardieBacker® cement board changed my studio practice. It's such a useful and efficient tool. This is where I roll or stamp texture into the clay and stiffen my slabs. The moist clay stretches across the cement board easily as I stamp and texture it. The board also absorbs moisture from the wet clay, making the slabs immediately workable. You do have to be careful, because some varieties absorb moisture too quickly, your slabs will dry out within minutes. HardieBacker® cement board (figure A), is used as underlayment behind tile and floors in kitchens and bathrooms to prevent moisture damage. Ceramic artists use it to handbuild on because it's lightweight, easy to cut, and doesn't contain plaster. Photo: Courtesy of James Hardie Building Products Inc.



the under side of the little leaf cup and make sure the lid fits the teapot (figure 7).

The Spout(s)

I have used templates to make spouts, but I really like to wing these so that I am looking at the personality of the pot I am working on, not the one I designed the template for. Make a bunch. Play around with fatness and length. What works with this form, with this texture, may not work with another. Play with multiple textures on the same pot. Look at placement for function. Imagine the water-level line in the pot and how it will move as the teapot is tipped. Make sure the spout is high enough that water won't come out of it when you fill the pot, but not too high that you have to almost flip the pot over to pour the tea out.

To make the spout, cut out a pentagon shape with one corner elongated. With the longest point facing toward you, fold the two topside edges together and join (figure 8). An optional dart cut from the middle of the pentagon forms a nice little paunch. After you've formed several spouts, hold them up to the body and determine which one you like best with the form (figure 9). Mark the outline of the spout at its eventual attachment place (figure 10), fill in the outlined area with multiple holes ¼ inch in



1 Assorted bisque-fired cylinders with carved texture.



2 Use several cylinders to roll a variety of texture into slabs.



3 Cut and drape the slab over a pipe to get the desired curve.



4 Join the curved slabs at each end and compress the join well.



5 Add leaf-shaped slabs to form the top and join them across the middle.



6 Cut a hole in the top of the slabs and add a textured ring for the collar.



7 Make lids from two small leaves and a narrow small cylinder.



8 Shape a variety of spouts cut from pen-tagonal shapes.



9
Try the spouts on for size and look and choose the best one for each teapot.



10
Mark where the spout will attach, cut holes and join it to the pot.



11
Flip pot gently, cut out a slab for the base, then slip and score each piece.



12
Attach the base to the pot and trim any excess clay around the exterior.



13
Gently lift and roll the pot to create a line at the base and to belly out the form.



14
Flip the pot and add four pinched knobs for feet. Let them firm up.



15
Once the feet are firm enough, flip the pot over and play with lugs and knobs.



16
Once you've found the best knobs and lugs for each teapot, attach them both.



17
Line the pot then dunk it in glaze up to the rim. Cover the spout so no glaze pours in.



18
Reclaimed garden tools, cut, turned, and drilled. Soon to be teapot handles.



19
Twiggy Cream and Sugar, mid-range red stoneware, with bone and teal glazes.

from the line, then score, slip and attach the spout.

The Bottom and Feet

At this point the base form should be just stiff enough to flip over, carefully. Use coils inside the form like caulking to strengthen any seams. The bottom of the teapot is yet another leaf. Mark the outline of the body on a slab, cut it out, then score and slip both edges and join (figures 11–12). Step back and decide whether the teapot needs feet. For footed teapots, I roll the base so the bottom curves out into a nice belly (figure 13). Doing this gives the teapot more volume both visually and literally. Let the pot set up a bit, upside down, before adding little feet—I take time for a tea break, to check emails, and to feed my Instagram addiction.

To make the feet, squish four balls of stiffer clay into little pyramids. Add the feet (figure 14), flip the pot over, and gently place the teapot back on the cement board to level it. The cement board will quickly absorb moisture out of the feet, making them firm enough to hold their shape.

Next, make a bunch of knobs of different shapes and sizes, and again, hold them up to the form to see which ones work best. Make the lugs out of short coils that narrow in the middle and curl them around the handle of a paintbrush. Place lugs and knobs on the teapot before attaching to ensure everything works together (figures 15–16). Attach components and make any small refinements such as sharpening a convex seam with a rib or a concave seam with a softly pointed tool. This highlights and compresses the seams, which prevents cracking.

Clean up any chunky bits and sharpen the spout for better pouring by trimming a sliver of clay from the lower edge of the spout's opening.

Dry, Fire, and Glaze

After the pot dries and has been bisque fired, line it with a food-safe glaze and wipe off any drips from outside. Let the liner glaze dry. Holding the teapot from the rim in one hand and covering the opening of the spout with the other hand, so no glaze pours in, dunk the teapot into glaze up to rim (figure 17). Add color to the lugs

and the knob to embellish if you wish. Wipe the feet bottoms clean. Simply dunking my pots in a single glaze allows the form and texture to do all the decorative work; the glaze just accentuates it. My glazes are predominantly satin in surface. The high-iron content in the clay bleeds into the glaze, especially in the thin areas where the glaze breaks over texture or sharp edges. The glaze emphasizes the texture and the satin surface speaks to the softness of the clay in its moist state. Subtle textures and lines can get lost in the reflections of shiny glazes. Sometimes I highlight the knob or lugs with a little extra color, further defining the different parts.

Reclaimed Wood Handles with Steel Wire

I make my handles out of reclaimed hardwood garden tools (figure 18) and 12-gauge steel ceiling-hanger wire. I cut up the broken wooden handles into various sizes, finish the ends with a router, drill holes through, then use a natural tung oil to finish the wood. These wooden handles feel soft and warm in the hand.

For the wire part, after I feed the steel wire through the hole, I bend it into shape with pliers, measuring the bend points with a ruler so each side of the handle is symmetrical. This wire is very stiff and hard to bend, but because it's designed to hang a ceiling, it's strong and will hold its shape well over time.



Sarah Pike makes pots in Fernie, British Columbia—a sweet little ski town in the Canadian Rocky Mountains, where it can get very snowy. She studied at the Alberta College of Art and Design, the University of Colorado, Boulder, and the University of Minnesota, Twin Cities. To see more, check out, www.sarahpikepottery.com, and follow her at www.facebook.com/sarahpikepottery, and Instagram @sarahpikepottery.